

Homework 2

1. The results of [minimax.py](#) playing against itself is shown in Figure 1. Since the minimax algorithm involves no stochasticity, the results are deterministic; each agent acts rationally to achieve the optimal outcome (5 dark, 11 light).

A small technicality: when researching Othello on the internet, the optimal Othello outcome with Minimax agents is 3-11 dark, not 5-11. This occurs because our specified game rules differ from standard Othello, which stipulates that the game ends when *neither* player has any moves. In the case where only one player has run out of moves, standard Othello continues with the move passing to the other player.

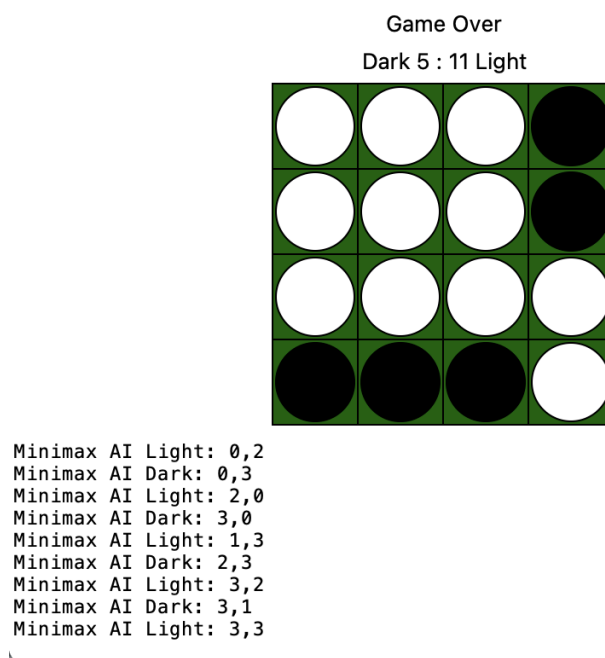


Figure 1.

2. When playing as dark, the minimax wins 8 out of 10 times, and while playing as light, minimax wins 10 out of 10 times. Research online for 4x4 Othello boards states that light has a guaranteed chance of winning when played optimally, which we see in Minimax's complete victory as light. While the Minimax agent is guaranteed to lose as dark if its opponent is playing optimally, a random agent allows the Minimax agent to seize victories, although it cannot do so all the time.
3. When playing as dark, minimax wins 0 out of 10 times, and while playing as light, minimax wins 10 out of 10 times. A rollout of 100 seems to give the MCTS algorithm the opportunity to explore most states of the 4x4 board before reaching a solution, so the results reflect that of optimal play.
4. MCTS beats Randy every time as both light and dark. While Randy just picks a uniformly random legal move each turn, MCTS runs 100 guided rollouts per move, allowing it to choose more optimal moves.